

FreeMat: A Free Alternative to MATLAB

FreeMat is an open source numerical computing environment and programming language that is similar to MATLAB and GNU Octave

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Available under the GPL license, FreeMat is a free environment that has been designed for “rapid engineering and scientific prototyping.” It is similar to commercial tools like Matrix Laboratory (MATLAB) from Mathworks and Interactive Data Language (IDL) from Research Systems Inc.

It also features codeless interface to external C, C++ and Fortran code, parallel-distributed algorithm development (via MPI), and some extended volume and 3D visualisation capabilities.

Tutorial to make a function

Here is a short tutorial to create combination and permutation functions for FreeMat, from the FreeMat project home page at code.google.com.

In order to create functions, you can use Notepad (Windows) or vi or gedit (Linux). FreeMat also comes with a built-in editor for this purpose (see fig 1).

To open the FreeMat editor, first run FreeMat and then, from the command line, type in the word ‘edit’ (without the quotes).

In FreeMat, a function is a script that starts with word ‘function’ and uses the following syntax:

```
function return_value =
function(parameter1,
parameter2,...,parameterN)
```

When the editor window opens up after typing ‘edit’, enter the following combination function:

```
function return_value=comb(n,r)
return_value=gamma(n+1)/(gamma
(n-r+1)*gamma(r+1));
```

To make it a function that FreeMat

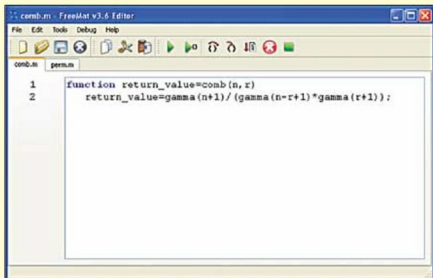


Fig. 1: Creating combination functions for FreeMat

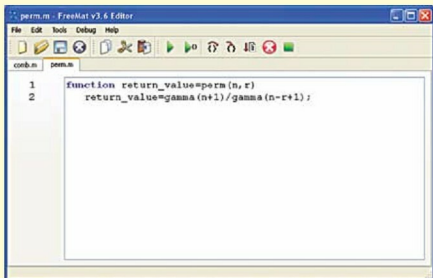


Fig. 2: Creating permutation functions for FreeMat

can use, store the file in a folder that is part of the FreeMat path. To check the path, use the path tool. Select Tools→Path Tool and check that the

directory where the function will be stored is part of the path.

To save the function, either select File→Save or click ‘Save’ icon. When